Examiner only

8. (a) (i) Explain how *multimode dispersion* arises in a glass fibre. [2]

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(ii) State what feature of a *monomode* fibre prevents multimode dispersion. [1]

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(b)

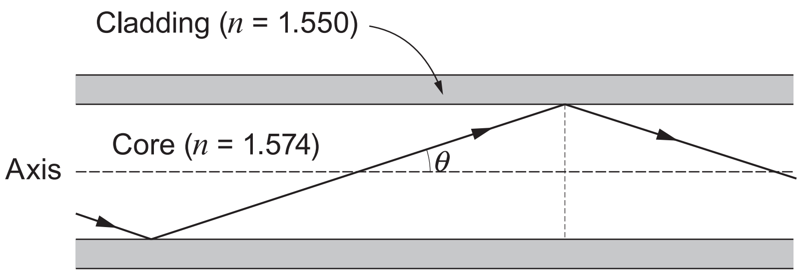

Diagram not drawn to scale

(i) Calculate the largest angle, θ, to the axis (see diagram) at which light can travel for long distances through the core of a multimode fibre, if the refractive index of the core is 1.574, and that of the cladding is 1.550. [3]

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Examiner only

(ii) Explain what would happen to light entering the fibre at an angle to the axis greater than the angle calculated in (b)(i). [2]

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END OF PAPER